PERFORMANCE SPECIFICATION SHEET

BATTERY, RECHARGEABLE, REPAIRABLE, NICKEL CADMIUM, VENTED, 24-VOLT, 10 AMPERE-HOUR, BB-432B/A

This specification is approved for use within Army CECOM Life Cycle Management Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-81757.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery shall be M81757/19-1.

2. Dimensions and weight. The dimensions and weight of the battery shall be as specified in 3.10.8 as modified herein and as shown on figure 1 herein. The weight of the battery shall be not greater than 34.0 pounds (15.4 kg).

3. Capacity. The minimum capacity of the battery shall be 10 ampere-hours (12.5 ampere-hours initially). When tested as specified in MIL-PRF-81757, the test parameters and requirements of Table I herein shall apply.

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TABLE I: Test parameters and requirements, BB-432B/A battery

Test	Test Method	Discharge Rate(A) 1/	Cut Off Voltage(V) 1/	Capacity (Ah) Minimum 1/
Initial capacity discharge	4.5.13	10	18	12.5
Humidity and Charge Retention	4.5.14	10	18	7.5
Shock	4.5.15	250	10	4.1
Vibration	4.5.16	250	10	4.1
Temperature Shock	4.5.17	10	18	10
Altitude	4.5.18	10	18	8.5
Twenty-second pulse discharge	4.5.20	250	10	4.1
Medium Rate Discharge and Operating Position	4.5.21	100	14.4	8.5
Cycle Life	4.5.23			
Normal Rate (Cycle 1)		10	18	10
Medium Rate		100	14.4	6.6
Normal Rate (Cycle 100)		10	18	10
Normal Rate (Cycle 200)		10	18	10

1/ Test parameters and requirements cited herein supersede those in the general specification.

4. Electrical interface requirements. The BB-432B/A battery shall have two connectors, located as described by Figure 1 herein:

a. Connector J1: a 2-contact MS3509-XX receptacle, qualified in accordance with specification MIL-PRF-18148
b. Connector J2: a 10-contact connector that meets the interface requirements of MS3474W12-10S

5. Cells. The BB-432B/A shall include 19 each cells, PIN M81757/1-2. Cells shall be connected inside the BB-432B/A battery using one wiring harness (see 6 below), seven M81757/3-2 conductors, six M81757/3-1 conductors, and three M81757/4-1 conductors. One M81757/3-4 conductor shall be used to connect the cells to the negative MS3509 terminal and one M81757/3-10 conductor shall be used to connect to its positive terminal.

6. Cell links. The cell links used for connecting cells in series shall be copper with a cross-sectional area not less than 0.031 in^2, with nickel plating that complies with SAE-AMS-QQ-N-290, Class 1, Grade E. Finish shall be smooth and free of corrosion, pits, and blow-holes.

7. Wiring harness. The BB-432B/A shall include 1 wiring harness, PIN M81757/20-1. Connector J2 shall be secured to the battery container with safety wire by means of a slotted, fillister-head screw and a nut complying with NASM21083-C3 (see Figure 1 for location). The screw, wire, and nut shall be made from corrosion resistant steel.

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8. MIL-PRF-81757 variance. The battery shall comply with MIL-PRF-81757 except as follows.

8.1 Modify the following paragraphs:

3.6.2 Fasteners. After the phrase "…free of any coating…" add the following: "except as otherwise specified herein".

3.7.4 Latches. Delete text in its entirety and replace with the following:

"The cover shall be secured to the container body by latches mounted on the container body, and latch hooks mounted on the cover. The latches and hooks shall be mounted on the cover and container body (see Figures 1 and 2 herein)."

3.7.5 Handles. Not applicable to this battery.

3.7.9 Connectors, intercell. This battery is designed for full maintenance; tamper-resistant fasteners shall not be used.

8.2 Add the following new paragraph:

"3.7.11.9 Shipping condition. BB-432B/A batteries shall be delivered complete with electrolyte to the proper level and completely discharged, but fully formed. Each battery shall be equipped with a shorting device across the output terminals of the J1 connector."

8.3 Add the following new paragraphs:

"3.8.4.3.1 Additional marking. In addition to the marking described by figure 3, the following marking shall appear on each battery:

WARNING

1. Area around battery must be well ventilated and cover removed during charge.

2. Do not allow caustic electrolyte to contact body, clothing, tools, etc. If contact is made wash immediately with water. Flush eyes with water only for 15 minutes and seek medical attention.

CAUTION
DO NOT USE ACID
ELECTROLYTE

Note: Use 0.17 ± 0.030 inch lettering for this marking

Acid will permanently damage this battery.
Avoid using tools which have been used on
acid type storage batteries. Consult technical
manual before servicing.

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Note: Use 0.09 ± 0.030 lettering for this marking

INSTRUCTION
1. Remove cover and brush off white carbonate deposits with vents in place. 2. Remove shorting plug or spring. 3. Refer to charging instructions in applicable technical manuals. 4. After charge, allow 1-2 hour stand, remove cell vents and adjust electrolyte using distilled water to 1/4 inches above plates for the cells. Replace cell vents and cover. 5. Periodically soak cell vents in distilled water.

Note: Use 0.09 ± 0.030 lettering for this marking"

"3.8.8 Warning Tags. Where shipping devices are furnished by the contractor to assure proper shorting, disallow cell leakage etc., alkali resistant tags with strong alkali resistant nonmetallic cord shall indicate action or actions to be taken by field personnel to remove and/or replace it."

8.4 Modify the following paragraphs:

3.10.20 Twenty-second pulse discharge. Delete "not less than 18.0 volts" from part (b) and replace with "not less than 10.0 volts."

3.10.22 Constant voltage discharge. This requirement is not applicable to this battery.

8.5 Add the following paragraph:

"3.10.31 Testing of wiring harness. When the wiring harness is subjected to either the wiring harness temperature test or to the electrolyte resistance test, the wiring harness shall exhibit the following:

a. There shall be no evidence of alkali migration into harness wires and components; breakdown of insulation; stripping of metal plating from connectors; corrosion of metal parts; nor cracking of potting compound.
b. The thermistor shall meet the requirements of MIL-PRF-81757/20(CR).
c. The thermoswitch shall close or open within the specified temperature ranges of MIL-PRF-81757/20(CR).
d. There shall be no mechanical failure of components.
e. There shall be no dimensional distortion beyond specified limits of MIL-PRF-81757/20(CR)."

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VERIFICATION

1. MIL-PRF-81757 variance. The battery shall comply with MIL-PRF-81757 except as follows.

1.1 Modify the following paragraphs:

4.2.1 Inspection of batteries.

a. Delete “tamper-resistant fasteners (see 3.7.9)” and replace with “two #8-32 UNC 3A screws, two spring washers, and two flat washers”

b. Add: “In addition, one wiring harness, PIN M81757/20-1, shall be provided.”

4.5.9 Handle strength test. Not applicable to this battery.

4.5.15 Shock test (basic design). Delete “Discharge the battery at the 9C-rate for 5 minutes. Measure the voltage of the battery at 5 minutes after the start of the discharge.” Replace with: “Discharge the battery at a constant current of 250 amperes in three twenty-second pulses with 120 seconds of rest between pulses.”

4.5.16 Vibration test. Delete in its entirety and replace with the following:

“The following test shall be performed. Upon completion, batteries or cells shall meet the requirements of 3.9.

“a. Remove cover from battery when necessary.

“b. Constant current charge each battery at the 1C rate.

“c. Secure cover to battery when necessary.

“d. Attach battery to vibration equipment in the normal, upright position.

“e. Connect battery to the test system.

“f. Attach no less than two accelerometers to battery in the direction of the vibration motion; use the spectral density average of accelerometers to control test levels.

“g. Subject battery to the vibration test of MIL-STD-810, Method 514, Procedure I (Operational service), using Annex D parameters for Category 14, Helicopters, vibration parameters for General Materiel, f_x values for Main Rotor of CH-47D.

“h. Disconnect battery from vibration equipment; attach battery to vibration equipment in a perpendicular axis.

“i. repeat steps e through h for a total of 3 mutually perpendicular axes.

“j. Each vibration period shall be not less than one hour.

“k. Discharge battery at the 0.1C rate during each vibration period; monitor battery voltage during discharge.

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"l. Upon completion of all three vibration periods, constant current charge battery at the 1C rate.

"m. Discharge the battery at a constant current of 250 amperes in three twenty-second pulses with 120 seconds of rest between pulses.

"n. Examine battery for the requirements of 3.9."

4.5.18 Altitude test. This test is required as described with the following exceptions:

a. For step b, use a test temperature of -1.1°C in lieu of -40°C

b. For step c, use a pressure that simulates 6,090 meters (19,980 ft.) of altitude in lieu of the 18,500 meters (60,700 ft.) pressure equivalent."

4.5.20 Twenty-second pulse discharge. This test is applicable to this battery. However: for step (b), stabilize at 27 ± 3°C (81°F) in lieu of -17.8 ± 1.1°C (0°F); for step (c), use a constant current of 250 amperes in lieu of the sloping discharge rate.

4.5.21 Medium rate, operation position. This test is applicable to the battery. However, the discharge rate shall be as specified in Table 1 herein in lieu of the 9C-rate specified in MIL-PRF-81757.

4.5.22 Constant voltage discharge test. This test is not applicable to this battery.

4.5.23 Cycling test. Delete text in its entirety and replace with the following:

"The following test shall be performed.

"a. Cycle 1. Charge at a constant current at the 1C rate to 1.55 ± 0.10 volts per cell, then 1.1 amperes for another three hours. Rest for one hour at open circuit. Then discharge at the normal rate specified herein to cut-off voltage.

"b. Cycle 2. Charge at a constant current at the 1C rate to 1.55 ± 0.10 volts per cell, then 1.1 amperes for another three hours. Rest for one hour at open circuit. Then discharge at the medium rate specified herein for 2.0 minutes.

"c. Cycles 3-98. Charge at a constant current at the 1.5C rate to 1.55 ± 0.10 volts per cell and rest for one hour at open circuit and discharge at the medium rate specified herein for 2.0 minutes.

"d. Cycle 99. Charge at a constant current at the 1.5C rate to 1.55 ± 0.10 volts per cell and rest for one hour at open circuit and discharge at the medium rate specified herein to the cut-off voltage specified.

"e. Cycle 100. Repeat step a.

"f. Cycle 101. Repeat step b.

"g. Cycle 102 -198. Repeat step c.

"h. Cycle 199. Repeat step d.

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"i. Cycle 200. Repeat step a.

"NOTE: Electrolyte level shall be checked after the 100th cycle. Add water if necessary. The amount of water added shall be recorded. Reconditioning is not allowed during the cycle life test."

4.5.27 Dielectric strength and insulation resistance test. This test is applicable with the following exception: Add the following test point: f. The battery terminals to the latches.

1.2 Add the following new paragraph:

"4.5.32 Wiring harness temperature test. The following test shall be performed. After completion of testing, the wiring harness shall meet the requirements of 3.1 and 3.10.31.

"a. Remove wiring harness from battery.

"b. Fixture the harness so that an ohmmeter is connected across the terminals of the thermistor and the S1 thermoswitch of the wiring harness.

"c. Place the harness into an environmental chamber; set the chamber to reach the first temperature set point in the table below. Once the set temperature has been reached, allow the test sample to stabilize for not less than 1 hour.

"d. Record resistance values for the thermistor and the thermoswitch.

"e. Repeat steps b and c for the second set point in the table below.

"f. Repeat step d until readings have been obtained through set point 4.

"g. Set the chamber to reach a temperature within a range of 56 ± 1.1°C (133°F); allow the test sample to stabilize for not less than 1 hour.

"h. Set the chamber to ramp up the temperature to not greater than 62.8°C (145°F) at a rate not greater than 1°C per five minutes while monitoring the continuity of thermoswitch S1 continuously.

"i. Once the thermoswitch closes, record the temperature and stop the temperature increase.

"j. Set the chamber to reach a temperature within a range of 50.6 ± 1.1°C (123°F). Allow the test sample to stabilize for not less than 1 hour.

"k. Set the chamber to ramp down the temperature to not less than 46°C (115°F) at a rate not greater than 1°C per five minutes while monitoring the continuity of thermoswitch S1 continuously.

"l. Once the thermoswitch opens, record the temperature and proceed to the next step.

"m. Return chamber to room ambient conditions. Allow the test sample to stabilize for not less than 1 hour. Remove the test sample and examine for the requirements of 3.10.31.

"n. Reinstall the wiring harness into the battery.

"Temperature Set Points, BB-432B/A Wiring Harness

(1) 10.0 ± 1.1°C (50°F)

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(2) 21.0 ± 1.1°C (70°F)
(3) 32.0 ± 1.1°C (90°F)
(4) 43.0 ± 1.1°C (110°F)”

1.3 Modify the Tables of MIL-PRF-81757 as follows:

Table II. Identification of defects. Add the following to Table II of MIL-PRF-81757:

Number	Description	Method of inspection
15	Incorrect thermistor values.	Ohmmeter
16	Date codes missing or incorrect or more than one date code on a battery.	Visual
17	Carrying strap on cover missing, twisted or torn. Carrying strap stitches are loose or broken, strap cut edges are not sealed to prevent unraveling.	Visual
18	Connector mounted incorrectly (key not oriented at 12 o'clock position).	Visual

Table IV. Qualification inspection of components and materials. Add the following to Table IV of MIL-PRF-81757:

Part	Examination and tests	Requirement paragraph	Test paragraph
Wiring harness	Wiring harness temperature test	3.10.31	4.5.32

Table VII, Group B inspection. Add the following to Table VII of MIL-PRF-81757:

Test No.	Examination and tests	Requirement paragraph	Test paragraph
I-4	Dielectric strength and insulation resistance (wiring harness only)	3.9, 3.10.27	4.5.27

Table VIII, Group C inspection. Add the following to Table VII of MIL-PRF-81757:

Test No.	Examination and tests	Requirement paragraph	Test paragraph
I-10	Electrolyte resistance (wiring harness only)	3.9, 3.10.3	4.5.3
II-8	Wiring harness temperature test	3.10.31	4.5.32 3

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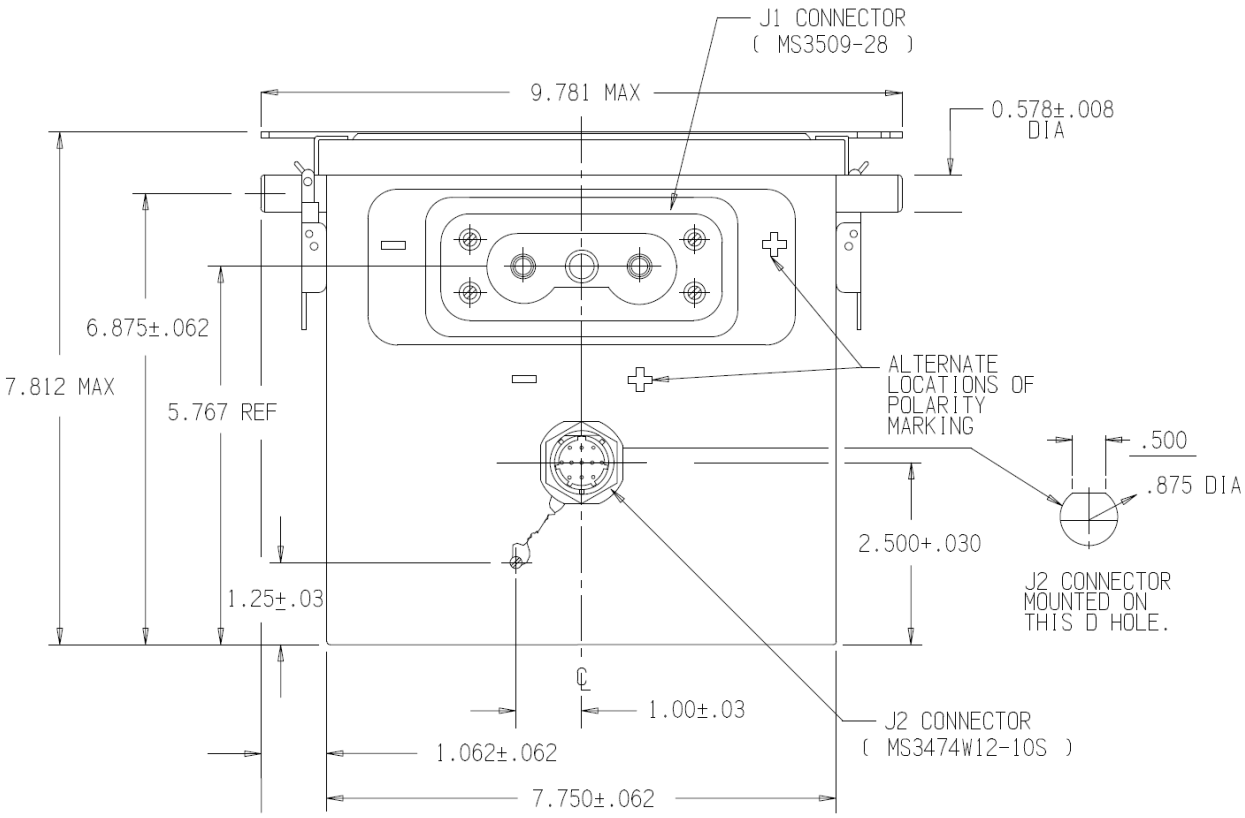


FIGURE 1. Battery Assembly BB-423B/A Front View

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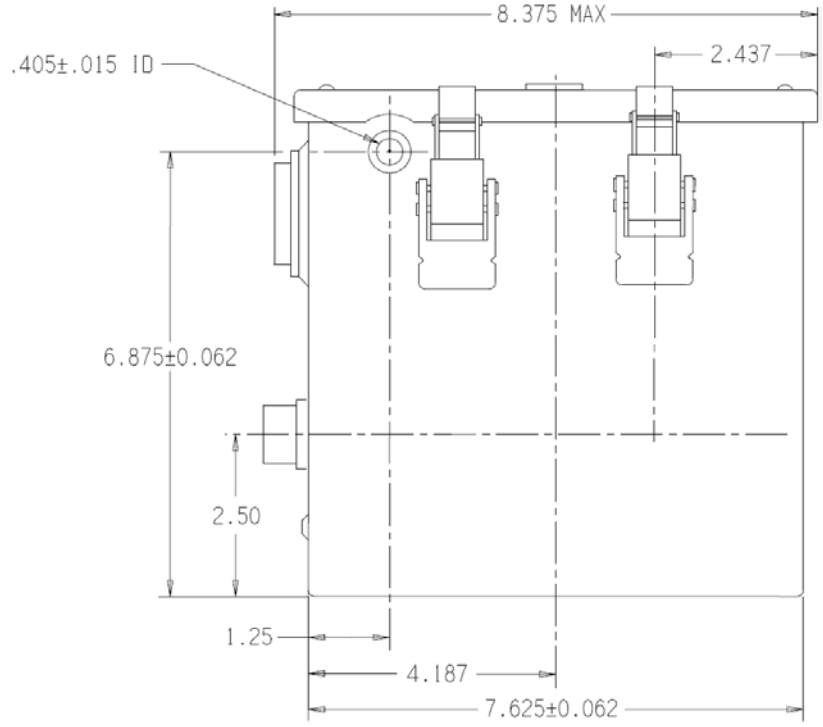


FIGURE 2. Battery Assembly BB-432B/A Side View

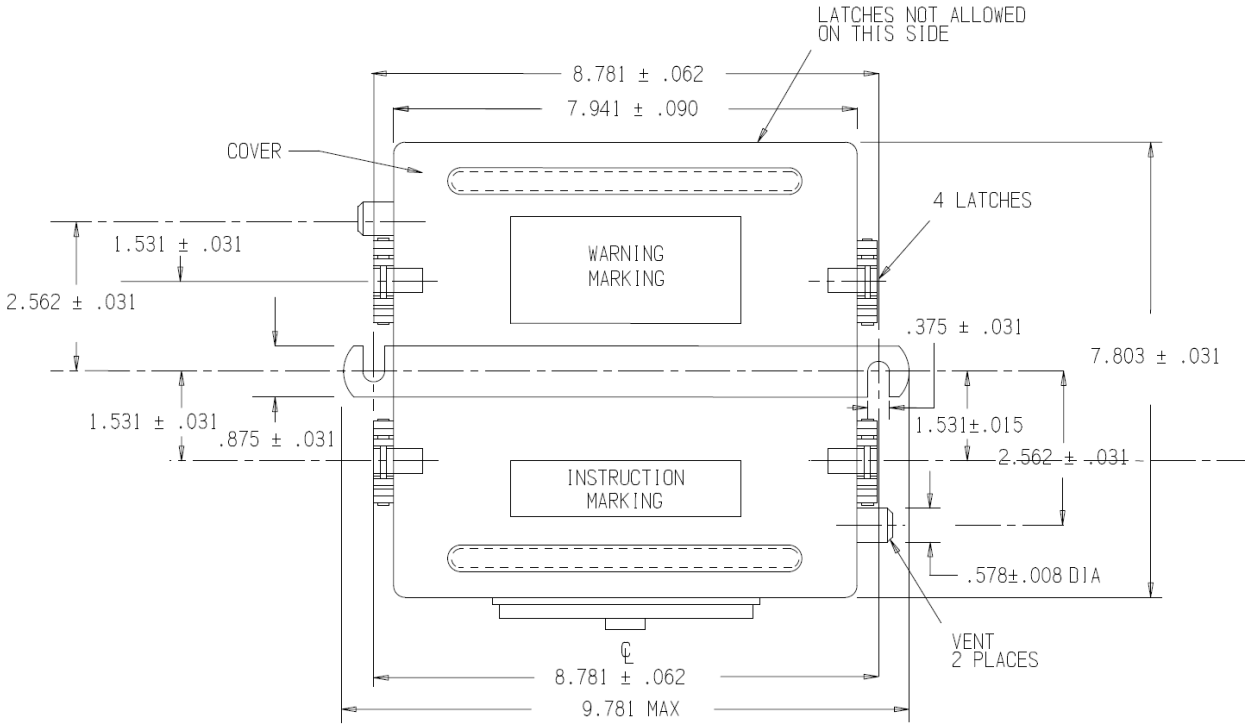


FIGURE 3. Battery Assembly BB-432B/A Top View

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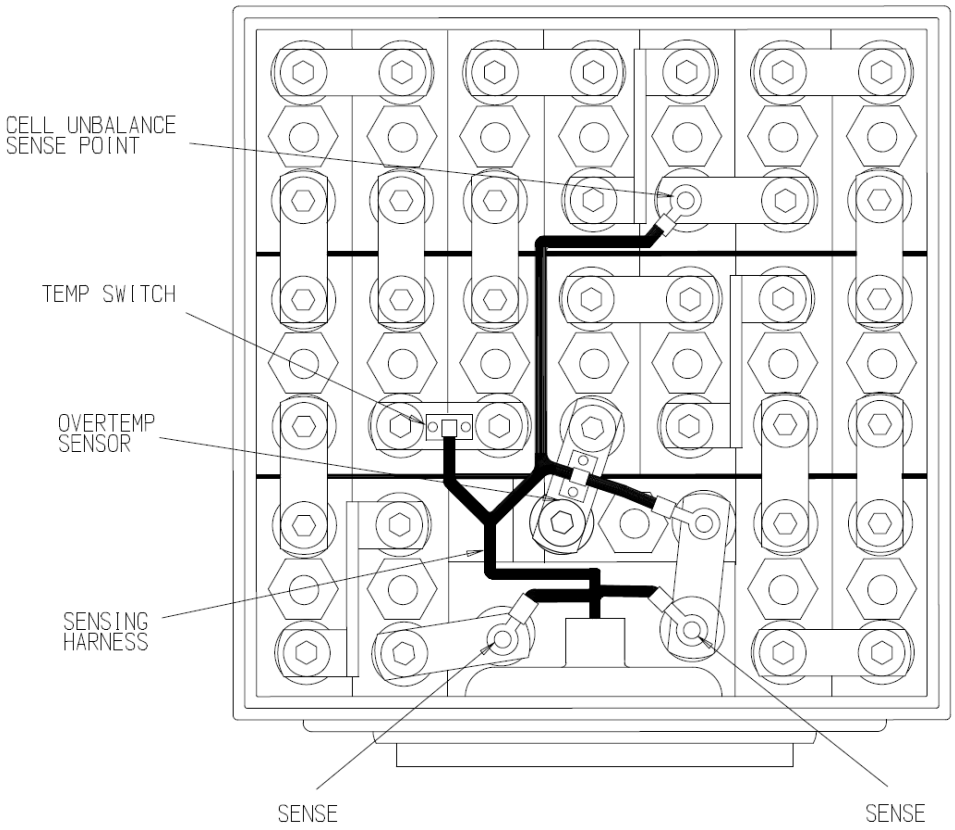


FIGURE 4: Cell Layout Inside BB-432B/A, Top View

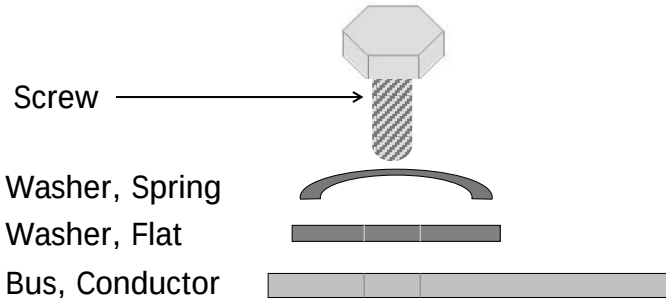


FIGURE 5: Screw/Washer Assembly Detail

FIGURE NOTES:

1. All dimensions are in inches, unless specified otherwise.

2. Vent tubes. The battery shall have two vent tubes shown in Figures 1, 2 and 3. The vent tubes shall be made of the same material as the battery case.

3. Holding strap. The holding strap shown in Figure 1 and Figure 3 shall be located on the centerline of the battery.

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4. Battery Container. The body and cover shall be made of the same alkali-resistant material. The dimensions of the container and the dimensions and locations of the receptacle mounting holes and hold down strap shall conform to Figures 1, 2, & 3.

5. Hardware. The following hardware is required to complete the cell assembly inside the BB-432B/A:

ITEM	NSN	P/N (CAGEC)	QUANTITY
Wiring Harness	6685-01-143-9075	M81757/20-1 (81349) 29817-001 (74025)	1
Bus, Conductor	6150-00-950-9325	M81757/3-2 (81349) 178A3227P23 (01002) 7861 (56770)	7
Bus, Conductor	6150-00-950-9324	M81757/3-1 (81349) 162A7812-25 (01002) 009337-002 (09052)	6
Bus, Conductor	---	M81757/3-4 (81349)	1
Bus, Conductor	---	M81757/4-1 (81349)	3
Bus, Conductor	---	M81757/3-10 (81349)	1
Screw, Hex-head, SS, #8-32 UNC 3A, 7/16”L			38
Washer, Spring	5310-00-118-1977	M81757/1-8A (81349) 16128-26 (74025)	38
Washer, Flat	5310-00-976-4621	162A7819-22 (01002)	38

Notes:
NSN = National Stock Number
P/N = Part Number
CAGEC = Commercial and Government Entity Code

Note. Potassium carbonate. Previous versions of specifications for the Army Chinook battery had requirements and test methods for potassium carbonate. It has been determined that failure to control and limit potassium carbonate in production would result in cells incapable of complying with the performance requirements herein. Users of this document should be aware that there remains a need to limit potassium carbonate levels in order to provide assurance of capability in both qualification and conformance inspections.

Custodians:
Army – CR

Preparing activity
Army – CR
(Project Number 6140-2009-020)

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